

OS_SKILL4

1. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
srivalli@DESKTOP-6S9T6SD: ~$ cd OSLab
srivalli@DESKTOP-6S9T6SD: ~/OSLab$ pwd
/home/srivalli/OSLab
srivalli@DESKTOP-6S9T6SD: ~/OSLab$ nano parser.c
srivalli@DESKTOP-6S9T6SD: ~/OSLab$ gcc parser.c -o parser
srivalli@DESKTOP-6S9T6SD: ~/OSLab$ ./parser
Enter a command/expression: x = 10 + 20;

--- TOKEN STREAM ---
Token: x           Type: IDENTIFIER
Token: =           Type: DELIMITER
Token: 10          Type: NUMBER
Token: +           Type: DELIMITER
Token: 20          Type: NUMBER
Token: ;           Type: DELIMITER

--- TOKEN VALIDATION ---
Token 'x' : Valid
Token '=' : Valid
Token '10' : Valid
Token '+' : Valid
Token '20' : Valid
Token ';' : Valid
```

--- PARSE TREE / EXECUTION STRUCTURE ---

COMMAND

```
|  
+-- x [IDENTIFIER]  
|  
+-- = [DELIMITER]  
|  
+-- 10 [NUMBER]  
|  
+-- + [DELIMITER]  
|  
+-- 20 [NUMBER]  
|  
+-- ; [DELIMITER]
```

Parsing completed successfully.

```
srivalli@DESKTOP-6S9T6SD: ~ × + ▾
srivalli@DESKTOP-6S9T6SD:~/OSLab$ ./parser
Enter a command/expression: x = 10 @ 20;
Lexical Error: Invalid character '@'

--- TOKEN STREAM ---
Token: x          Type: IDENTIFIER
Token: =          Type: DELIMITER
Token: 10         Type: NUMBER
Token: 20         Type: NUMBER
Token: ;          Type: DELIMITER

--- TOKEN VALIDATION ---
Token 'x' : Valid
Token '=' : Valid
Token '10' : Valid
Token '20' : Valid
Token ';' : Valid

--- PARSE TREE / EXECUTION STRUCTURE ---
COMMAND
|
+-- x [IDENTIFIER]
|
+-- = [DELIMITER]
|
+-- 10 [NUMBER]
|
+-- 20 [NUMBER]
|
+-- ; [DELIMITER]

Parsing completed successfully.
```

```
srivalli@DESKTOP-6S9T6SD:~/OSLab$ ./parser
Enter a command/expression:

--- TOKEN STREAM ---

--- TOKEN VALIDATION ---
Empty command detected.

--- PARSE TREE / EXECUTION STRUCTURE ---
No parse tree generated for empty command.

Parsing completed successfully.
srivalli@DESKTOP-6S9T6SD:~/OSLab$
```